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Ensemble learning with RAG model to reduce redundant question topics in auto-generated exam questions

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Abstract

Reducing redundant question topics during automatic question generation (AQG) is essential for enhancing the quality of test sheets in assessment. Existing AQG models frequently generate repetitive questions due to insufficient named entity (Question Topic) diversity. This study aims to reduce redundancy in auto generated questions by improving diversity in question topics. The methodology is organised in three main phases: first, a fuzzy ontology mapping technique with ensemble learning is applied to generate expanded entity-relationship set generation from external Knowledge Graphs. Second, the generated entity-relationship set is integrated with Retrieval Augmented Generation (RAG) model for source text expansion via augmentation. Third, T5-shd, a pretrained AQG model is adopted to reduce repetition in generated questions. Comparison against baselines such as T5-e2e and T5-ppl shows substantial performance gain as well as reduction in redundancy. Experimental results on various datasets show that the proposed RAG+T5 model reduces redundant question topics along with an improvement in terms of ROUGE-2 metric (up to 5%) and BERTScore (up to 12%) over existing methods. The application of Ensemble Pruning, specifically with the m-EPIC algorithm, further enhances accuracy while reducing computational overhead (around 26%). These findings highlight the efficacy of combining ensemble learning with RAG-based transformers to refine AQG, ensuring improved question diversity and balanced relevance of generated questions. Additionally, this approach helps to reduce model complexity in Automatic Question Generation.

Keywords Named entity selection, Ensemble pruning algorithm, Knowledge graphs, Retrieval augmented generation, Automatic question generation.

1 Introduction

In educational assessments, Automatic Question Generation (AQG) is widely used to create exam questions, quizzes, and practice tests [1–7]. However, one of the critical challenges in AQG is the generation of redundant question topics, which reduces the effectiveness of assessments and student engagement. Traditional AQG models, which rely on templates, neural networks, or language models, often fail to generate diverse

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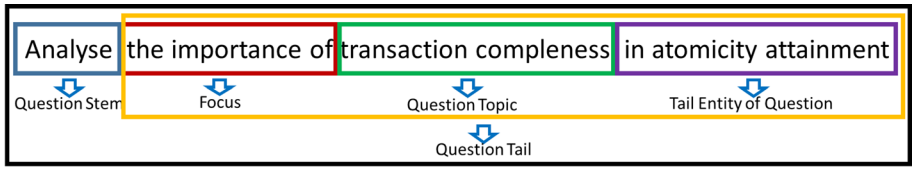


Fig. 1 Components of an Exam Question

Table 1 Need for diversity of question topic terms (entities) in AQG

Source Text	Naive Questions	Requirements	Expected Questions
A transaction in a database is a unit of work with ACID properties (Atomicity, Consistency, Isolation, and Durability). The transaction log records and ensures the durability of changes made during transactions, aiding in recovery and maintaining data consistency.	Define a transaction in a database.	Diversity of key entities (question topic terms)	Discuss the role of atomicity in normalisation
	List the properties in transaction management.		Describe the role of the transaction log in transaction management

questions [4–7] due to their limited ability to expand named entities within a given subject domain.

To overcome this limitation, incorporating strategies that promote diversity in question formulation by dynamically modifying the question topic within the question tail component (Fig. 1) is essential. Consider the source text: "A database transaction is a unit of work that is executed as a single operation. It groups one or more operations on a database to be executed as a single unit. It follows ACID properties to ensure that data remains consistent, even in cases of system failure or concurrent execution." The questions with the same question topic term may be generated as given below: (1) What is a database transaction? (2) Write a short note on database transaction. (3) Explain the importance of ACID properties in database transactions. The redundancy in question topic selection must be addressed by expanding the source text to yield and generate better questions such as: (1) How does isolation affect concurrent transactions? (2) Explain the importance of ACID properties in database transactions. (3) Discuss the role of durability to ensure recovery in database systems. Here, balance between the quantity and distinctness of questions in test sheets is highly essential (as shown in Table 1) to address the gap. While AQG can generate expected number of questions, care must be given to ensure that they are relevant as well as non-redundant.

Existing AQG systems have attempted to address redundancy through various techniques. Some approaches focus on template-based question generation to ensure structural consistency, while others incorporate neural networks to improve contextual understanding. Additionally, Knowledge Graph-based AQG models have been introduced to enhance entity retrieval for relevant question topic term classification [1, 2, 5, 8]. Despite these advancements, the issue of generating questions with redundant question topics remains unresolved. There is still a need for a robust mechanism that integrates external knowledge to balance relevance and distinctness of topic terms in auto-generated questions.

Building on this foundation, this study introduces a hybrid approach that integrates ensemble learning with Retrieval Augmented Generation (RAG) to mitigate redundancy

and enhance entity-relationship pair (graph) extraction. The proposed method incorporates structured ontology integration to extract relevant entity-relationships from external knowledge graphs and transform it into augmented source text using the RAG model [9, 10]. This augmented source text is processed using pretrained Transformer (such as the T5 model) for AQG to ensure diverse range of question topics in auto-generated questions. The ensemble learning [11–14] employed to refine the entity set, reduces overlap in extracted topics while preserving their relevance. This method ensures that the generated questions are not only contextually rich but also balanced in terms of uniqueness and subject relevance, expanding the scope of possible question topics.

The primary contributions of this study include:

1. Designing the Ontology with Retrieval Augmented Generative model to improve topic distinctness in auto-generated questions.
2. Investigating Entity Retrieval models for extracting distinct Question Topics via expanded entity set generation.
3. Applying Ensemble Learning technique to refine overlap in extracted topics while preserving their relevance.

The remainder of this paper is organized as follows: Sect. 2 presents a review of related work, discussing existing AQG models and their limitations in handling redundant question topics. Section 3 describes the research methodology, detailing the integration of ontology-based retrieval, ensemble learning, and the RAG model. Section 4 outlines the experimental setup, including datasets, evaluation metrics, and baseline comparisons. Section 5 discusses the results, analyzing the effectiveness of the proposed approach in reducing redundancy and improving topic diversity. Finally, Sect. 6 provides conclusion and future research directions.

2 Related work

Automatic Question Generation (AQG) has evolved through various methodologies (Table 2), including template-based, neural network-driven, knowledge representation-based, and entity-guided approaches [4–8]. Regardless of the approach, three key components must be considered: (1) question tail selection, (2) question stem selection, and (3) question supplement or pattern selection [2–4]. Template-based AQG methods ensure structural consistency but often lack adaptability, while neural network-based models improve contextual understanding but struggle with out-of-vocabulary (OOV) cases and question pattern variations. Knowledge representation (KR)-based [4, 8, 15]

Table 2 AQG workflows in the literature

AQG Approaches	Methodology Adopted	Remarks
Template AQG [3, 15]	Named Entity Extraction with syntactic or semantic scoring to fit the templates	Models to generate newer question patterns are required
Neural Network AQG [14, 24]	Masked Language Models based text to question transformation using attention mechanism, maxout pointers etc.	Low interpretability to domain specific question patterns requires improvement
KR based AQG [7, 15, 24]	Vocabulary based Embedding vector representation or Deep semantic parsing based source text representation	Limited degree of generalisability to varying text spans needs improvement
Entity context AQG [14, 25, 26]	Chunk scoring based source to target question generation	Representational uncertainty with respect to semantic labelling must be addressed

and entity-guided [16–19] approaches attempt to address these limitations by leveraging semantic analysis and domain knowledge. However, KR-based methods face challenges with diverse text structures, and entity-guided models require external knowledge models to resolve vocabulary constraints and linguistic uncertainties [19–23]. Addressing these limitations requires refining of improved entity set generation strategies which remains a key research challenge.

Several studies emphasize the role of domain ontologies and knowledge models in AQG [11–13, 20–22, 27–29]. Ontology-based methods, combined with similarity analysis or logic-based reasoning, improve interpretability and entity alignment. However, existing techniques often prioritize individual entity matches over contextual relevance, which may extract unique terms which are irrelevant to the context. Additionally, complex mapping systems employ various similarity metrics, such as frequency-based indexing and semantic proximity measures. But mitigating false negatives due to the integration of techniques need attention. Most of these methods tend to either emphasize relevance, leading to repetitive question topics, or prioritize uniqueness at the expense of contextual relevance [4, 8]. In this direction, AQG systems have started incorporating ensemble models.

Recent advancements in ensemble learning have demonstrated its potential (Table 3) to improve various generative tasks by reducing redundancy and refining interpretability [27–29]. Techniques such as bagging and majority voting have been widely used, but overfitting issues persist due to the rapid expansion of decision trees. To address these issues, pruning methods, including 'Minimum Individual Error Pruning', 'Complementariness Measure-based Pruning' and "Ensemble Pruning via Individual Contribution Ordering" were designed to iteratively refine decision subsets [11, 12, 28] to balance model accuracy with reduced complexity. However, maintaining question topic diversity is an essential research direction. To build more precise and context-aware AQG frameworks, ontology-driven entity or question topic selection needs more attention.

3 Methodology

The proposed architecture (Fig. 2) integrates external knowledge model-mediated extraction of entity-relationship pairs which is followed by source text augmentation using the Retrieval Augmented Generation (RAG) framework. First, the entity-relation pairs for key entity or question topic terms extraction phase is considered. The second phase focuses on augmented text generation based on generated entity-relationship set (from previous phase) which is taken as input by the baseline T5 model.

Table 3 Study on algorithms for ensemble pruning

Ensemble Pruning Algorithms	Methodology Adopted	Remarks
Minimum Individual Error Pruning [11, 13]	Selection of top-k decision stumps in non-increasing order of accuracy	Determination of subforests size is challenging
Reduce-Error Pruning [12, 13, 28]	Stepwise exclusion of decision stumps with higher disagreement against the incremental subforest	Stepwise exclusion of decision stumps on k-fold data is quite challenging
Complementariness Measure [11, 28]	Stepwise decision stump exclusion using entropy measure or gini index against the incremental subforest	Stepwise exclusion of decision stumps on k-fold data is still challenging
Ensemble Pruning via Individual Contribution Ordering [12, 28]	Determining the orientation ordering of decision stumps with four vital cases	Can be implemented to improve the performance of Named Entity Expansion process

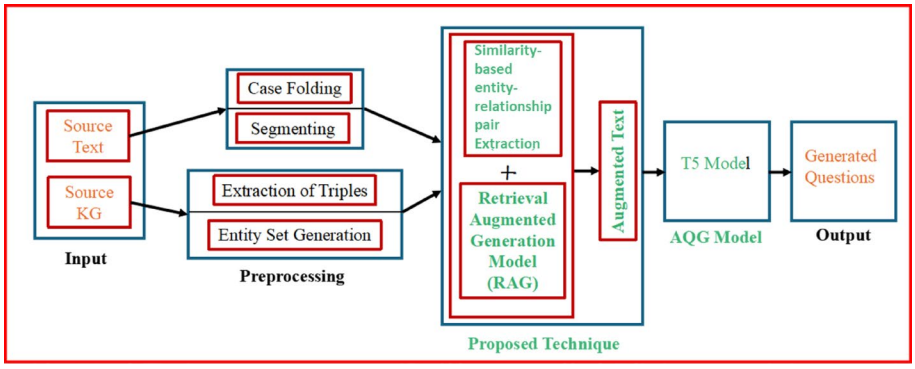


Fig. 2 Architecture of the Proposed Work

3.1 Data preprocessing and entity extraction

Input text, consisting of ill-formed questions or segmented source text, is preprocessed using Natural Language Processing (NLP) techniques such as tokenization, case folding, and entity extraction. Entity categorization is performed using the IsimScore metric (Eq. 1), which calculates bigram relevance and semantic relationships. This step facilitates Ontology-mediated entity-relationship generation during the retrieval of relevant domain-specific knowledge.

$$ISimScore = \frac{\prod_{POS=NN} Sim(Ei+1\,|Ei,\ Ej+1|\,Ej)}{\sum_{i=1}^{n}\sum_{j=1}^{m} Sim(Ei,\ Ej)} \tag{1}$$

Where, POS = Parts of Speech Tag. Sim = Similarity between unigram/bigram entities. Ei + 1, Ei = Entities extracted from input text. Ej + 1|Ej = Entities extracted from knowledge graphs.

3.2 Ensemble learning for Entity-relationship pair generation

Ontology-mediated named entity selection techniques are essential for selecting and ranking entities in context. However, independently curated domain ontologies exhibit inconsistencies, making entity selection complex. To address this, ensemble modeling is proposed (Fig. 3). The approach iteratively combines multiple base learning models using bootstrapping and aggregation techniques. This bagging-based method constructs a Random Forest (RF) model by selecting random samples from source data (D) and the feature set (F). The integration of weighted aggregation techniques refines entity-relationship set generation, maintaining contextual accuracy. It uses methods such as selective weighting, equal weighting or averaging. The subset selection process follows the assumption that using 63% of source data records and log(F) of the feature set generates feasible Random Subsets (RS) (Eq. 2).

$$RS = \frac{D * F}{0.63\,(D) * \log\,(F)} \tag{2}$$

When using Random Subsets (RS) to create decision trees as base learners, there is a risk of generating too many sub-forests, up to a maximum of 2(RS-1). This results in overfitting of the model or overly complex computational predictions. To address this issue, the goal is to select the decision stump with the highest prediction accuracy as given in Eq. 3.

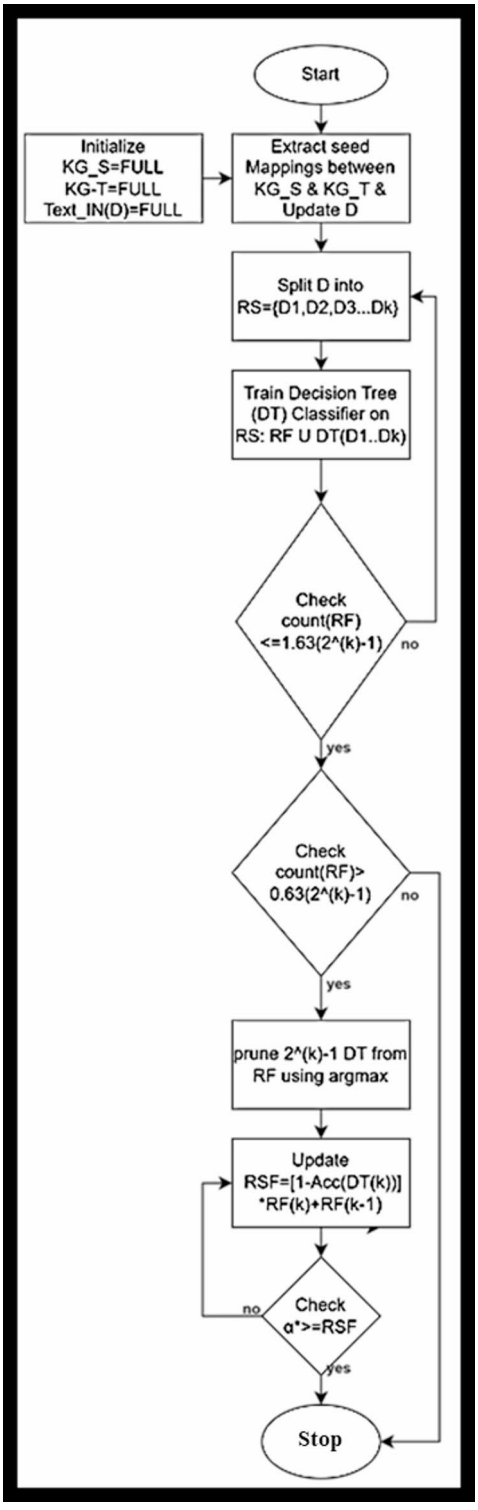


Fig. 3 Ensemble Learning Flowchart

$$\mathrm{RF} = \sum argmax\,[Acc\,(DT)] \tag{3}$$

The linear growth of decision trees (DT) within Random Subsets Forest (RSF) is done by choosing the decision tree (DT) with the highest accuracy, denoted as Acc(DT), and adding it to the RSF in a cumulative manner, as described in Eqs. 4 and 5 respectively.

$$\mathrm{RSF} = [1 - \mathrm{Acc}\,(\mathrm{DT}\,(\mathrm{k}))]\,{}^{*}\mathrm{RF}\,(\mathrm{k}) + RF(k-1) \tag{4}$$

$$\mathrm{RSF}\,(\mathrm{k}) = \alpha\,{}^{*}o'\;[DT\,(k)] \tag{5}$$

The constant α represents the classification loss at instance k. When triangular inequality is applied, it is noted that the classification loss of the Random Subsets Forest (RSF) on D is not greater than the classification loss of RSF at each instance, k in D (Eq. 6). This enhances the performance of the proposed ensemble model.

$$\alpha * (\mathrm{o}\,[\mathrm{k}] - \mathrm{o}'\,[\mathrm{DT}\,(\mathrm{k})])^2 \geq \;\; [[1 - \mathrm{Acc}\,(\mathrm{DT}\,(\mathrm{k}))] * \mathrm{RF}\,(\mathrm{k}) + \mathrm{RF}\,(\mathrm{k}-1)]^2 \tag{6}$$

3.3 Ensemble pruning to reduce overpopulation

Handling the overpopulation of ensemble models is an important research area aimed at reducing overfitting challenges. Many forest pruning algorithms [24–26] have been developed with the goal of achieving both individual model accuracy and diversity to reduce overall model complexity. Here, it's crucial to prune decision trees that are unlikely to contribute effectively.

This model architecture (Fig. 4) shows the structured approach to prune decision trees (stumps) of Random Forest (RF) to manage overpopulation and to mitigate overfitting. The process begins with generating ensemble of decision trees (Random Forest) from the split data chunks of preprocessed texts, where extracting decision stumps—simplified decision trees is followed. The model enhances performance by incorporating the update mechanism consisting of two key functions: Argmax population and Argmin Pruning. The Argmax population function selects the most relevant decision stumps, while the Argmin Pruning function removes those which are negligible in their contribution. By

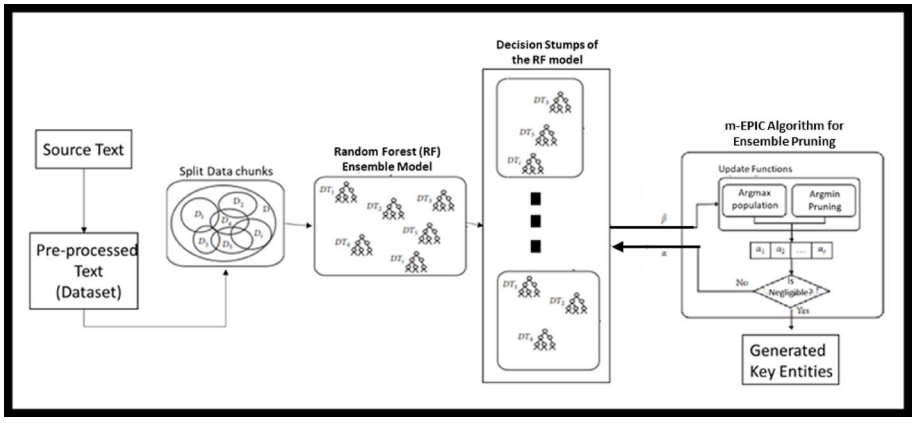


Fig. 4 Ensemble Model Generation and Pruning Process

implementing these update functions, the approach aligns with the broader goal of optimizing ensemble models by selectively refining their structure, thereby reducing computational complexity and improving generalization.

Several heuristics-based pruning techniques, such as Kappa Pruning and Minimum Individual Error Pruning, struggle to select diverse yet accurate decision stumps. Likewise, algorithms like Complementary Measure and Improved Orientation Ordering iteratively expand subforests by selecting trees based on the highest diversity or disagreement using the Gini index. While these methods evaluate individual Decision Stump (DS) accuracy and incorporate the uth decision stump into the subforest with k-v stumps at instance v, they prune k stumps with indices up to u – 1, as shown in Eqs. 7 and 8. However, for this entity relation pair extraction task, EPIC, which is among the best-performing models, is further refined with four update rules, introducing the m-EPIC pruning technique, as detailed in Algorithm 1.

$$RSF\,(v) = argmax \sum\nolimits_{v=1}^{DS} (DS\,(u) + \sum\nolimits_{t=1}^{u-1} h\,[RSF\,(v)]) + RSF\,(v-1) \tag{7}$$

$$RSF\,(v) = argmax \sum\nolimits_{v=1}^{DS} (DS\,(u) + \sum\nolimits_{t=1}^{u-1} y \oplus\ h\,[RSF\,(v)] + y \oplus\ RSF(v-1)) \tag{8}$$

Where, RSF(v) and RSF(v-1) refer to the random subforests at instances v and v-1; DS(u) stands for the uth accurate decision stump in v^{th} iteration. y and h[...] are the ground truth class labels and heuristics to attain diversity-accuracy trade off through XOR operation on orientation driven predictions. To further optimize ensemble selection, the modified "Ensemble Pruning via Individual Contribution Ordering (EPIC)" algorithm is proposed (Algorithm 1). This method prioritizes decision stumps that enhance model interpretability while maintaining high prediction accuracy by defining four new EPIC update rules (Algorithm 1). Here, excluding decision stumps is highly required based on higher disagreement for incremental sub-forests (Eq. 9). Also, the limited computation of orientation relationships between decision stumps and the random forest, is taken care in the modified EPIC (m-EPIC) algorithm.

$$RSF\,(v) = \frac{[DS\,(c) \oplus\ RSF\,(c)] + [DS\,(w) \oplus\ RSF\,(c)] + [DS\,(c) \oplus\ RSF\,(w)] + [DS\,(w) \oplus\ RSF\,(w)]}{N} \tag{9}$$

Where, N represents the total number of data records taken for training each DS; RSF(c) and RSF(w) represent correctly and wrongly classifying Random Sub Forests; DS(c) and DS(w) represent correctly and wrongly classifying Decision Stumps of each Random Sub Forest. By integrating ensemble learning with Retrieval Augmented Generation, this methodology enhances topic distinctness to minimize redundancy in auto-generated questions.

Algorithm 1 Modified EPIC Pruning

Inputs: Source data D, Feature set F, Classification Loss Constant α.
Output: Ensemble Model Random Forest RSF.

```
// Random Subsets Generation
RS = (D * F) / (0.63(D) * log(F))
// Ensemble Model Generation (Random Forest RSF)
RSF = ∅
For each instance k in 1 to RS:
  Train a Decision Tree DT(k) on a random subset of the source data D
  Acc(DT(k)) = Accuracy of DT(k) on the training data
  RSF(k) = α * (ok - o'[DT(k)])^2
  RSF = [1 - Acc(DT(k))] * RSF + RSF(k-1)
end for
// Modified EPIC update rule
For each v in 1 to RS:
  RSF(w) = argmax(∑_(v=1)^DS[y⊕RSF(v-1)])
  RSF(c) = argmin(∑_(v=1)^DS[y⊕RSF(v-1)])
  DS(c) = argmax(∑_(t=1)^u[y⊕h[DS(v)]])
  DS(w) = argmin(∑_(t=1)^u[y⊕h[DS(v)]])
  RSF(v) = ([DS(c)⊕RSF(c)] + [DS(w)⊕RSF(c)] + [DS(c)⊕RSF(w)] + [DS(w)⊕RSF(w)])
/ N
  if (RSF(v) < 0.6)   //Pruning Constraint
    RSF = RSF - RSF(v)
  end if
end for
Return RSF
End for
```

4 Experiments, results and discussion

4.1 Dataset description and experimental setup

In the proposed work, several open-source domain ontologies like ODT [30], DCO [31], DPO [32], MAMO [33], and ODM [34] are utilized in experiments to generate subset Ontology Mappings for question topics selection (Named Entity Expansion) in the AQG process. A detailed description of these ontologies is presented in Table 4. MAMO represents mathematical models, their attributes, and variables formally. ODM records metadata for semantically annotating data mining scenarios. ODT, originally part of ODM, is now independently implemented as a taxonomy of data types. DPO covers models and modeling parameters in the Data Mining Domain. DCO validates incoming data and represents processes in data analysis. Secondly, the datasets used to analyze the outcome of the proposed work (i.e.), the AQG task. Different Question-answer pair datasets [2, 8] such as LearningQ-7 K, BCLS-1k, Allen why-how-3k etc., are processed to conduct experiments and to analyze the performance. A multi-stage stratified sampling is carried out with these datasets and around 1.6k QA pairs are selected and organized accordingly. Models such as KeyBERT, TNER etc., are incorporated for entity

Table 4 Data (Knowledge) model description

Sl No	Dataset	Classes	Properties	Max. Depth	Max Child nodes	Avg Child nodes
1	MAMO	105	3	5	12	2
2	DPO	541	220	13	23	3
3	DCO	73	38	7	5	2
4	ODM	482	39	6	21	2
5	ODT	406	17	10	22	3

extraction from 1.6k instances. BSNER models are applied for boundary smoothing [1, 15, 17], which is taken for manual curation through sufficient entity markers in highlight (HL) mode. The Scikit-Learn, Pandas, Numpy libraries, and Matplotlib are used for Ensemble Learning and Pruning. The parameter settings used in experimental setup is detailed in Table 5.

4.2 Performance analysis– proposed AQG model vs. baselines

An exploratory study is conducted on the proposed Random Forest model with Retrieval Augmented Generation and T5 Transformer to assess the model performance on AQG task. The study is done by fine-tuning T5 Transformer (end-to end (T5-e2e), pipelined(T5-ppl) and shared(T5-shd)) on the augmented text, generated using the RAG model from generated entity-relationship pairs. Results are compared against baselines using n-gram relevance metrics (ROUGE-2, ROUGE-L) and semantic relevance metric (BERTScore). The fine-tuned model outperforms its baselines, achieving a performance improvement (Table 6) of approximately 5% in ROUGE-2 and 12% in BERTScore. The successful validation on the complete test dataset, without skipping any sample, confirms the reduction of redundant terms in question topic generation.

4.3 Results and performance analysis– expanded entity set generation models

The results of Expanded Entity Set Generation to balance distinctness and relevance in question topic selection are shown in Table 7. Using Random Forest (RF) models with different pruning algorithms—MIEP, REP, CMP, and m-EPIC, the RF models eliminate redundant or less informative entities. Among them, RF + m-EPIC achieved the best performance, likely due to its ability to prune unnecessary expansions from external knowledge sources. RF + CMP and RF + REP also improved the count of selecting informative entities, however falls back in terms of accuracy. These findings suggest that

Table 5 Parameter settings used in experimental setup

Learning Models	Parameter	Value/Setting
Random Forest	Number of Trees	100
	Max Depth	5
	Criterion	Gini Index
	Feature Subset Selection	$\log_2(f)$ per split
Ensemble Pruning (m-EPIC)	Diversity Threshold	0.6
	Max Ensemble Size (after pruning)	~47% of total trees
	Diversity Metric	XOR-based disagreement in orientation ordering
Entity Extraction	Model	KeyBERT, BSNER
	N-gram Range	(1, 3)
	Confidence Threshold (BSNER)	0.85
RAG Configuration	Top-k Retrieved Docs	5
	Token Limit for Context	128
T5 Transformer	Variants Used	T5-ppl, T5-e2e, T5-shd
	Learning Rate	3.4e-5
	Batch Size	32
	Epochs	4
	Max Sequence Length	128 tokens
Hardware Used	GPU	Tesla T4/P100 (Colab-pro environment)
	RAM	12–25 GB (based on session)

Table 6 Performance comparison of LLM baseline vs. hybrid models for question generation

Datasets/Models	HotpotQA-90k			CWQ-4k			SciQ-13k			BCLQ-2k		
	R–2	*R*-L	BS	*R*–2	*R*-L	BS	*R*–2	*R*-L	BS	*R*–2	*R*-L	BS
Baseline Models												
T5-e2e [1]	0.40	0.38	0.95	0.41	0.39	0.96	0.41	0.39	0.97	0.41	0.39	0.77
T5-shd	0.41	0.38	0.71	0.42	0.39	0.72	0.42	0.38	0.74	0.41	0.39	0.73
T5-ppl	0.41	0.39	0.74	0.42	0.40	0.74	0.42	0.40	0.77	0.42	0.40	0.76
RAG + T5 Hybrid Model ($T5_{v2}$)												
$T5_{v2}$-e2e [1]	0.42	0.39	0.94	0.43	0.40	0.94	0.43	0.40	0.93	0.42	0.40	0.75
$T5_{v2}$-shd [23]	0.43	0.40	0.74	0.44	0.40	0.75	0.43	0.40	0.75	0.43	0.41	0.74
$T5_{v2}$-ppl [23]	0.43	0.40	0.77	0.44	0.41	0.78	0.44	0.41	0.80	0.44	0.40	0.79
RF + RAG + T5 Hybrid Model (Proposed) ($T5_{v3}$)												
$T5_{v3}$-e2e	0.44	0.41	0.92	0.45	0.43	0.93	0.45	0.43	0.91	0.45	0.43	0.74
$T5_{v3}$-shd	0.43	0.41	0.75	0.44	0.41	0.73	0.43	0.41	0.72	0.45	0.43	0.74
$T5_{v3}$-ppl (proposed)	0.45	0.42	0.88	0.46	0.43	0.89	0.46	0.43	0.90	0.46	0.43	0.89
EPIC + RF + RAG + T5 Hybrid Model (Proposed) ($T5_{v3}$)												
$T5_{v3}$-e2e	0.46	0.44	0.95	0.45	0.43	0.44	0.45	0.43	0.93	0.44	0.41	0.83
$T5_{v3}$-shd	0.45	0.43	0.78	0.45	0.41	0.43	0.44	0.41	0.73	0.43	0.41	0.85
$T5_{v3}$-ppl (proposed)	0.47	0.44	0.90	0.46	0.43	0.45	0.46	0.43	0.89	0.45	0.42	0.89

In the above table, R-2, R-L and BS represents ROUGE-2, ROUGE-L and BERTScore respectively

combining RF + m-EPIC can attain both diversity and improved relevance in question topic selection.

4.4 Analysis with ensemble pruning algorithms on different datasets

An empirical study was conducted to evaluate different Ensemble Pruning algorithms in addressing data overfitting in the proposed Random Forest model. The objective was to optimize weighted triple voting scores while reducing model overpopulation. The performance of various pruning techniques was assessed using the Ensemble Size (ES) metric, which measures the ratio of decision nodes in the pruned model to those in the aggregated model. Results (Table 8) indicate that the modifed EPIC algorithm outperforms others in entity-relationship pair generation, while compared to alternative approaches. These findings demonstrate that despite similar performance scores the EPIC achieves improved accuracy (around 2%) with a reduced ES (around 26%), effectively minimizing model overfitting.

4.5 Performance analysis of T5 baseline models vs. hybrid models for question generation

The performance analysis compares baseline and hybrid models for question generation using metrics such as ROUGE-2 (R-2), ROUGE-L (R-L), and BERTScore (BS) across four datasets: HotpotQA-90k, CWQ-4k, SciQ-13k, and BCLQ-2k. Baseline models (T5-e2e, T5-shd, T5-ppl) focus only on relevance but are outpaced by hybrid models in terms of balancing relevance and distinctness. Table 9 discusses the time complexity of components in the proposed AQG framework.

- *Baseline Models*: T5-e2e achieves R-2 scores of up to 0.41, with high BS scores around 0.97. T5-shd and T5-ppl slightly improve on R-2 and R-L scores but show variability in BS, particularly lower for T5-shd.
- *RAG + T5 (T5v2)*: Improves R-2 to 0.43 and R-L to 0.41, while maintaining BS around 0.80.
- *RF + RAG + T5 (T5v3)*: Further enhances R-2 to 0.46, and shows strong R-L and BS scores, with a notable BS score of 0.94.
- *m-EPIC + RF + RAG + T5 (T5v3)*: Offers the best overall performance, with R-2 reaching 0.47, and BS consistently high across all datasets.

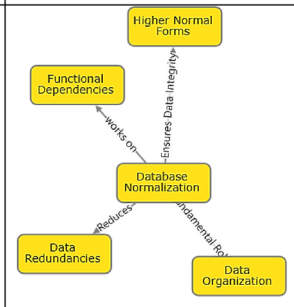
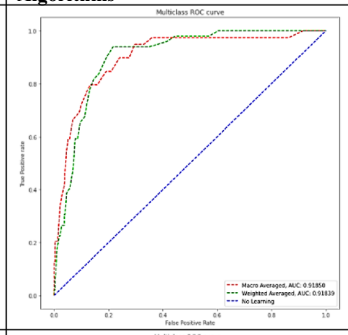
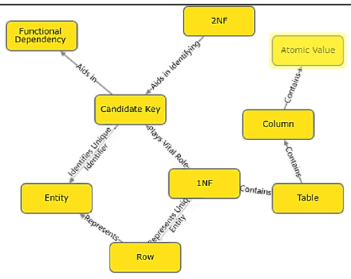
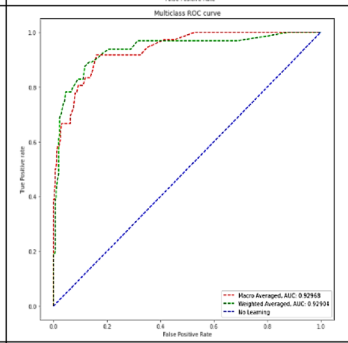
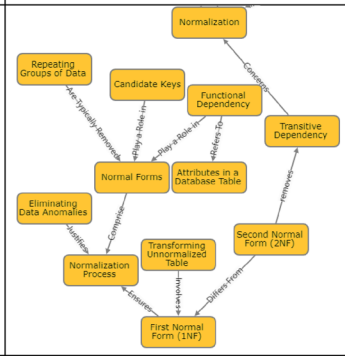
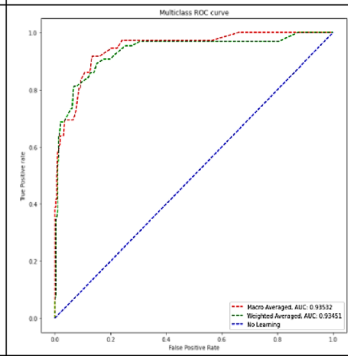
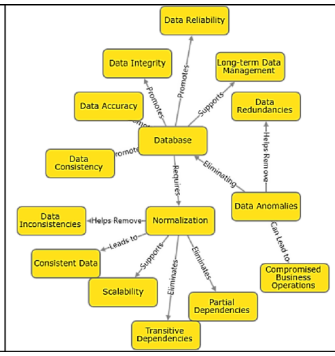
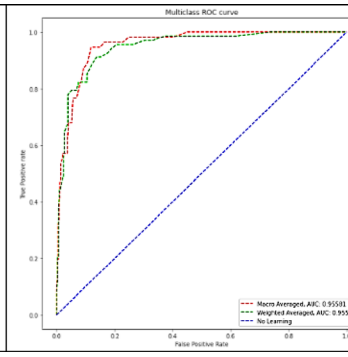
The progression from baseline to increasingly complex hybrid models demonstrates significant improvements, particularly in semantic relevance, as shown by BS and R-L scores. The proposed EPIC + RF + RAG + T5 model represents the most improved variant for balancing redundancy with enhanced question topic term generation and relevance in auto-generated exam questions, excelling all metrics.

4.6 Positive and negative results of entity expansion and AQG

The performance of the pipelined T5-base-RAG-AQG model is significantly influenced by the quality of the Knowledge Graph (KG) used for entity expansion. Analysis of the proposed multi-task learning model (EPIC-OM-RAG-AQG) reveals that using an inappropriate KG introduces irrelevant entity expansions, reducing diversity and negatively impacting results (Table 10, second entry). From this, two key observations emerge: (1) entity-relationship extraction must be precise, and (2) redundancy in relationships should be minimized. While the proposed model outperforms its baselines in KG-based

Table 7 Results and performance comparison of entity-relationship graph extraction

Source Text: Higher normal forms in DBMS offer advantages like improved data integrity and reduced redundancy. Candidate keys in 1NF identify unique attributes, while in 2NF, they determine functional dependencies. Eliminating data anomalies in normalization ensures data consistency and accuracy. Functional dependency defines relationships between attributes. 1NF focuses on atomicity, while 2NF removes non-key functional dependencies.

Pruning Algorithm	Results of RF with pruning Algorithms	Performance scores of RF with pruning Algorithms
RF+MIEP		
RF+REP		
RF+CMP		
RF+EPIC		

"RF– Random Forest, MIEP– Minimum Individual Error Pruning, CMP– Complimentariness Measure, REP - Reduced Error Pruning, EPIC– Ensemble Pruning via Individual Contribution ordering"

Table 8 Performance Comparison - Ensemble pruning algorithms

Datasets	Performance of Pruning Algorithms							
	MIEP		REP		CMP		Modified EPIC	
	Acc	ES	Acc	ES	Acc	ES	Acc	ES
DPO	0.94	0.74	0.94	0.63	0.95	0.59	0.95	0.47
DCO + ODT	0.92	0.79	0.93	0.65	0.93	0.62	0.93	0.53
ODM	0.92	0.73	0.94	0.62	0.93	0.58	0.93	0.49
MAMO	0.93	0.72	0.93	0.60	0.93	0.58	0.94	0.45

Table 9 Time complexity of learning tasks in the AQG framework

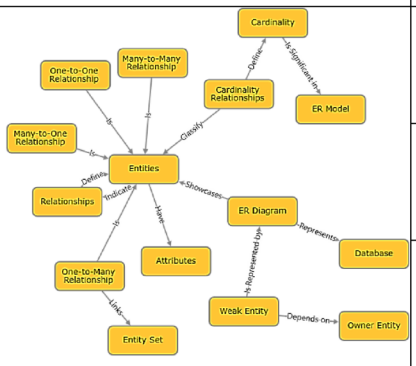
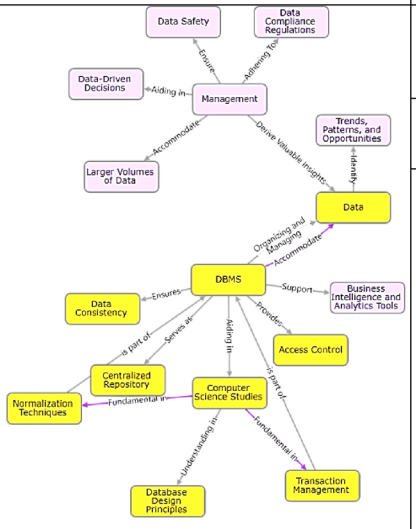
Learning Task	Description	Approximate Time Complexity	Description
Random Forest Construction	Ensemble of decision stumps using bootstrapped samples	$O(k \cdot n \cdot \log f)$	k: trees, n: instances, f: features
Ensemble Pruning (m-EPIC Algorithm)	Pruning based on diversity and disagreement using four update rules	$O(k^2)$	k: trees, Pairwise diversity measures across sub-forests
Ontology-Based Entity Extraction	Entity-relationship pair generation using similarity metrics (e.g., lsimScore)	O(m)	m: ontology terms (linear in practice with curated ontologies)
Retrieval-Augmented Generation (RAG)	Document retrieval + sequence generation	$O(k \cdot l)$	k: retrieved docs, l: token length
T5 Transformer Fine-tuned Inference	Generation from augmented text	$O(s \cdot d)$	s: sequence length, d: decoder's diversity

entity-relationship extraction, data cleaning remains crucial area, as poor data quality can significantly affect model performance.

5 Conclusion and future work

This study presents an advanced approach to Automatic Question Generation (AQG) by integrating ensemble learning with Retrieval-Augmented Generation (RAG) through a Random Forest-based entity expansion model. By utilizing ontology-based entity-relationship extraction and refining entity-relationship extraction using ensemble pruning techniques, the model ensures a balanced trade-off between relevance and distinctness in question generation. Experimental results demonstrate that the fine-tuned T5 model surpasses baseline approaches (T5-e2e, T5-ppl, and T5-shd) across various evaluation metrics, achieving a notable improvement in ROUGE-2 (up to 5%) and BERTScore (up to 12%). The application of Ensemble Pruning, specifically with the m-EPIC algorithm, further enhances accuracy while reducing computational overhead (around 26%). Comparative evaluations indicate that the hybrid EPIC + RF + RAG + T5 model outperforms existing approaches, effectively addressing the challenges associated with redundancy and topic relevance in AQG. Despite the promising improvements, certain limitations due to the data noise in ontological data or inappropriate KG sources as crucial areas for further research.".

Table 10 Positive vs. Negative results (Entity-relationship graph extraction and AQG)

Source Text	Naive Questions	Expansion with Entity Graph	Generated Questions
Cardinality relationships in an ER Diagram define the number of instances one entity can be associated with another. They include one-to-one, where each entity has a unique association; one-to-many, where one entity connects to multiple others; many-to-one, where multiple entities link to a single one; and many-to-many, where multiple entities relate to several others.	Discuss the significance of cardinality in the ER Model.		Classify the cardinality relationships between entities in an ER Diagram.
	Define one-to-many relationship in ER Diagram.		Explain the concept of a one-to-many relationship in an ER Diagram.
	What is the purpose of cardinality in the ER Model?		Illustrate how a weak entity is denoted in an ER Diagram.
A DBMS serves as a centralized repository, organizing and managing data to ensure data integrity and consistency. Understanding database design principles, normalization techniques, and transaction management is fundamental in creating robust database systems. The use of appropriate data types, indexing, and defining primary keys and foreign keys are key elements in database design.	List down data compliance regulations		Enumerate trends, patterns and opportunities identified by data
	Define Data safety		Illustrate that data management ensures data safety
	Explain transaction management in the context of computer science studies		Compare the fundamentals of computer science studies in transaction management and normalisation techniques

Acknowledgements
This research was carried out at E-Learning and HCI Lab, Department of Computer Applications, National Institute of Technology, Tiruchirappalli.

Author contributions
Mr. R. Tharaniya sairaj has conceptualized the study, developed the methodology, performed the experiments, and wrote the initial draft of the manuscript. Dr. S. R. Balasundaram provided feedback to refine the approach, involved in the review and editing process and supervised the research work.

Funding
No funding was received for the implementation of the above work.

Data availability
The datasets generated during and/or analysed during the current study are available in the NCBO BioPortal repository, https://bioportal.bioontology.org/ontologies/.

Declarations

Ethics approval and consent to participate
Ethics approval and consent to participate were not applicable as this research did not involve human or animal participants.

Consent for publication
Consent for publication is not applicable as this manuscript does not include data from individuals.

Competing interests
The authors declare no competing interests.

Received: 19 November 2024 / Accepted: 11 July 2025
Published online: 01 August 2025

References

1. Sychev O, Anikin A, Prokudin A. Automatic grading and hinting in open-ended text questions. Cogn Syst Res. 2020;59:264–72.
2. Das B, Majumder M, Sekh AA, Phadikar S. Automatic question generation and answer assessment for subjective examination. Cogn Syst Res. 2022;72:14–22.
3. Kusuma SF, Siahaan DO, Fatichah C. Automatic question generation with various difficulty levels based on knowledge ontology using a query template. Knowl Based Syst. 2022;249:108906.
4. Cong QBG, Li C, Sun A, Chen H. An evaluation of classification models for question topic categorization. J Am Soc Inform Sci Technol. 2012;63(11):2220–33. https://doi.org/10.1002/asi.22611.
5. Huang P-C, Chan Y-H, Yang C-Y, Chen H-Y, Fan Y-C. EQGG: automatic question group generation. IEEE Trans Learn Technol. 2024;17:1994–2007. https://doi.org/10.1109/TLT.2024.3430225.
6. Tomikawa Y, Suzuki A, Uto M. Adaptive question–answer generation with difficulty control using item response theory and pretrained transformer models. IEEE Trans Learn Techno. 2025;17:2186–98. https://doi.org/10.1109/TLT.2024.3491801
7. Chung H-L, Chan Y-H, Fan Y-C. Handover QG: question generation by decoder fusion and reinforcement learning. IEEE/ACM Trans Audio Speech Lang Process. 2024;32:3644–55. https://doi.org/10.1109/TASLP.2024.3426292.
8. Willis A, Davis G, Ruan S, Manoharan L, Landay J, Brunskill E. Key phrase extraction for generating educational question-answer pairs. In: Proceedings of the 2019 ACM SIGCHI Conference on Human Factors in Computing Systems. 2019. pp. 1–10. https://doi.org/10.1145/3330430.3333636.
9. Lewis P, Perez E, Piktus A, Petroni F, Karpukhin V, Goyal N, Kiela D. Retrieval-augmented generation for knowledge-intensive nlp tasks. Adv Neural Inform Process Syst. 2020;33:9459–74.
10. Anand A, Goel A, Hira M, Buldeo S, Kumar J, Verma A, Gupta R, Shah RR. SciPhyRAG-Retrieval augmentation to improve LLMs on physics Q&A. In: International Conference on Big Data Analytics. 2023. pp. 50–63.
11. George G, Lal AM. Review of ontology-based recommender systems in e-learning. Comput Educ. 2019;142:103642.
12. Cardoso SD, Silveira D, M., Pruski C. Construction and exploitation of an historical knowledge graph to deal with the evolution of ontologies. Knowl Based Syst. 2020;194:105508.
13. Jianrui, Yang L. Subgraph-aware virtual node matching graph attention network for entity alignment. Expert Syst Appl. 2023;231:120694. https://doi.org/10.1016/j.eswa.2023.120694.
14. Zhang X, Zhang J, Fu X. A viewpoint adaptation ensemble contrastive learning framework for vessel type recognition with limited data. Expert Syst Appl. 2024;238:122191. https://doi.org/10.1016/j.eswa.2023.122191.
15. Sun K, Yu J, Li J, Hou L. Exploring sequence-to-sequence taxonomy expansion via Language model probing. Expert Syst Appl. 2023;122321.
16. Dataset MAMO. https://bioportal.bioontology.org/ontologies/MAMO. Accessed 24 Nov 2023.
17. Dataset ODT. https://bioportal.bioontology.org/ontologies/ONTODT. Accessed 24 Nov 2023.
18. Dataset DCO. https://bioportal.bioontology.org/ontologies/GDCO. Accessed 24 Nov 2023.
19. Saha S, Saha M, Mukherjee K, Arabameri A, Ngo PTT, Paul GC. Predicting the deforestation probability using the binary logistic regression, random forest, ensemble rotational forest, reptree: A case study at the Gumani river basin, India. Sci Total Environ. 2020;730:139197.
20. Assi A, Dhifli W. Instance matching in knowledge graphs through random walks and semantics. Future Generation Comput Syst. 2021;123:73–84.
21. Amador-Dominguez E, Serrano E, Manrique D, Hohenecker P, Lukasiewicz T. An ontology-based deep learning approach for triple classification with out-of-knowledge-base entities. Inf Sci. 2021;564:85–102.
22. Osman I, Yahia SB, Diallo G. Ontology integration: approaches and challenging issues. Inform Fusion. 2021;71:38–63.
23. Raffel C, Shazeer N, Roberts A, Lee K, Narang S, Matena M, Liu PJ. Exploring the limits of transfer learning with a unified text-to-text transformer. J Mach Learn Res. 2020;21(140):1–67.
24. Jiang F, Yu X, Du J, Gong D, Zhang Y, Peng Y. Ensemble learning based on approximate reducts and bootstrap sampling. Inf Sci. 2021;547:797–813. https://doi.org/10.1016/j.ins.2020.08.069.
25. Khalifa FA, Abdelkader HM, Elsaid AH. An analysis of ensemble pruning methods under the explanation of random forest. Inform Syst. 2024;120:102310. https://doi.org/10.1016/j.is.2023.102310.
26. Jiang F, Hu Q, Yang Z, Liu J, Du J. A neighborhood rough sets-based ensemble method, with application to software fault prediction. Expert Syst Appl. 2025;264:125919. https://doi.org/10.1016/j.eswa.2024.125919.

27. Rodríguez-Revello J, Barba-González C, Rybinski M, Navas-Delgado I. KNIT: ontology reusability through knowledge graph exploration. Expert Syst Appl. 2023;228:120239.
28. Sarailidis G, Wagener T, Pianosi F. Integrating scientific knowledge into machine learning using interactive decision trees. Comput Geosci. 2023;170:105248.
29. Alobaid A, Garijo D, Poveda-Villalón M, Santana-Perez I, Fernández-Izquierdo A, Corcho O. Automating ontology engineering support activities with ontoology. J Web Semant. 2019;57:100472.
30. Pham BT, Prakash I, Singh SK, Shirzadi A, Shahabi H, Bui DT. Landslide susceptibility modeling using reduced error pruning trees and different ensemble techniques: hybrid machine learning approaches. CATENA. 2019;175:203–18.
31. Iorio C, Aria M, D'Ambrosio A, Siciliano R. Informative trees by visual pruning. Expert Syst Appl. 2019;127:228–40.
32. Dataset DPO. https://bioportal.bioontology.org/ontologies/ONL-DP. Accessed 24 Nov 2023.
33. Garcia ACB, Vivacqua AS. Grounding knowledge acquisition with ontology explanation: A case study. J Web Semant. 2019;57:100487.
34. Dataset ODM. https://bioportal.bioontology.org/ontologies/ONTODM-CORE. Accessed 24 Nov 2023.

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